CLASS 9 — SCIENCE · FORMULA SHEET

Chapter 10: **Tissues in Action** | Quick revision, one page

Plant tissues

- Meristematic: apical (tips of root/shoot, length), lateral (cambium, girth), intercalary (base of leaves/internodes, regrowth).
- Simple permanent: parenchyma (storage, photosynthesis as chlorenchyma, floating as aerenchyma), collenchyma (flexibility, support), sclerenchyma (dead, lignified, strength; husk of coconut).
- Protective: epidermis (stomata, root hairs, waxy cuticle) and cork (dead cells with suberin) in the bark.

Complex permanent tissues

- Xylem: tracheids, vessels, xylem parenchyma, fibres; conducts water and minerals upward; dead cells.
- Phloem: sieve tubes, companion cells, phloem parenchyma, fibres; conducts food (both directions); living cells (except fibres).

Animal tissues

- Epithelial: squamous (flat, lining of mouth, skin), cuboidal (kidney tubules), columnar (intestine), ciliated; stratified for protection.
- Connective: blood and lymph (fluid), bone (rigid, skeleton), cartilage (smooth, ear, nose, joints), areolar, adipose (fat storage), tendon (muscle to bone), ligament (bone to bone).

Muscular and nervous tissues

- Skeletal: striated, voluntary, long, multinucleate. Smooth: unstriated, involuntary, spindle-shaped, single nucleus. Cardiac: striated, involuntary, branched, single nucleus.
- Nervous tissue: neurons with cell body, dendrites and axon; carries impulses; the brain, spinal cord and nerves.

Organisation

- Cells → tissue → organ → organ system → organism.
- Stomata: pores in the epidermis of leaves bordered by two bean-shaped guard cells; they exchange gases and lose water vapour (transpiration).

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